

we are working on a stress, mood, burnout personal informatics application. these are the user goals we wanted to support:



Here is the structured personal informatics (PI) framework mapping your users' targeted actions to a robust, multi-source data inventory.

1. Inventory of Actions

To ensure clear tracking, each action is categorized and assigned a unique code (A1 through A7), aligning directly with your four core user goals.

- **A1: 3-Day Horizon Check** | Check scheduled tasks to see what is due in the next 3 days. *(Goal: Maximize efficiency / Reduce cognitive friction)*
- **A2: 9:00 PM Digital Wind-Down** | Start a 9:00 PM cutoff to quiet down notifications and help wind down. *(Goal: Balanced lifestyle / Effective coping)*
- **A3: Post-Requirement Doom-Scroll Lock** | Create a 30-minute lockdown on doom-scrolling apps immediately after completing requirements. *(Goal: Reduce decision fatigue / Balanced lifestyle)*
- **A4: Physical Workspace Boundary** | Physically close the laptop, clear the desk, and place work materials out of sight at a fixed time daily. *(Goal: Balanced lifestyle / Effective coping)*
- **A5: After-Hours Academic Blackout** | Use automated app blockers to lock out school portals, academic apps, and communication channels after hours. *(Goal: Reduce decision fatigue / Balanced lifestyle)*
- **A6: Identical Task Batching** | Group identical task types (e.g., admin or heavy writing) into isolated time blocks. *(Goal: Maximize efficiency / Reduce cognitive friction)*
- **A7: 5-Minute Friction Buster** | Force yourself to open a document and work for exactly 5 minutes without judging output. *(Goal: Maximize efficiency / Reduce cognitive friction)*

2. Inventory of Data

To demonstrate that actions are being taken *and* goals are being reached, the app requires a mix of passive device logging, ecosystem integrations, and brief self-reported metrics (Ecological Momentary Assessments).

Digital Footprints & System Logs (Passive)

- **D01: App Usage Duration:** Total active screen time broken down by app category (Productive, Social Media, Communication).
- **D02: App/Screen Open Timestamp:** Exact time a specific app, portal, or designated screen is brought to the foreground.
- **D03: App/Website Access Denied Count:** The number of times a user attempts to open an app while a blocker/lockdown is active.
- **D04: System Notification State:** The system-level status of the device (e.g., Do Not Disturb active, Muted, or Default).
- **D05: Muted Notification Count:** The total volume of notifications intercepted and held during a wind-down or blackout block.

Ecosystem & Environment Data (Integrations/Sensors)

- **D06: Task Deadline Proximity Delta:** The total number of tasks flagged as "Due within 72 Hours" fetched via API integration with the user's task manager.
- **D07: Task Completion Timestamp:** The exact time a task status changes to "Completed" in the app or integrated ecosystem.
- **D08: Device Hardware State Log:** OS-level telemetry capturing when a laptop lid is physically closed or a desktop goes into deep sleep.
- **D09: Continuous Input Activity Duration:** Time spent typing or actively using shortcut commands without pauses longer than 15 seconds.

- **D10: Character/Word Count Delta:** The numerical volume of text added to a file during an active focus window.
- **D11: Calendar Block Categorization:** The alignment score comparing scheduled calendar blocks against actual app usage categories during that time window.

Subjective Logs & Biometrics (Outcomes/Goal Attainment)

- **D12: Pre-Action Urge/Friction Score:** A rapid, 1-to-5 in-app slider capturing subjective task resistance or the urge to procrastinate.
- **D13: Post-Action Focus & Clarity Score:** A rapid, 1-to-5 in-app slider evaluating perceived mental clarity or efficiency directly following an action.
- **D14: Wearable Recovery Metrics:** Passive sleep onset latency and sleep quality index pulled from third-party wearable APIs (e.g., Apple Health, Oura).
- **D15: Daily Burnout & Stress Index:** A brief, end-of-day self-assessment measuring perceived emotional exhaustion and lifestyle balance.

3. Matrix of Actions to Data

The matrix below maps the relationship between actions and data. Per your requirements, data points overlap across multiple actions to create a cohesive data model, and **every single action maps to exactly 5 distinct data points** to validate both compliance (action taken) and outcome (goal reached).

Action Code	Data Points Involved	How Data Validates the Action & Goal
A1: 3-Day Horizon Check	D01 , D02 , D06 , D12 , D13	Triggers when the task screen is opened (D02) and viewed (D01) alongside upcoming work loads (D06). Goal achievement (reduced cognitive friction) is proven by checking the drop from pre-check anxiety (D12) to post-check clarity (D13).
A2: 9:00 PM Wind-Down	D01 , D04 , D05 , D14 , D15	Validates compliance via DND states (D04) and muted alerts (D05) alongside drops in late-night screen time (D01). Goal achievement (recovery/balance) is proven via better sleep metrics (D14) and lowered daily burnout (D15).
A3: Doom-Scroll Lock	D01 , D03 , D07 , D12 , D13	Instantly triggered by a task completion (D07). Tracks lockdown efficacy via blocked app attempts (D03) and 0-minute usage (D01). Goal tracking assesses if blocking the app breaks the dopamine loop urge (D12) and yields evening clarity (D13).
A4: Workspace Boundary	D01 , D08 , D13 , D14 , D15	Confirms action via absolute hardware closure (D08) followed by zero device usage (D01). Goal attainment is verified by tracking a boost in evening mental detachment (D13), sleep scores (D14), and lower overall burnout (D15).

Action Code	Data Points Involved	How Data Validates the Action & Goal
A5: Academic Blackout	D01 , D03 , D04 , D12 , D15	Tracks automated enforcement via active block states (D04), blocked attempts to cheat the system (D03), and actual zero-use patterns (D01). Goal tracking maps whether curbing after-hours access reduces decision fatigue (D12) and long-term burnout (D15).
A6: Identical Task Batching	D01 , D07 , D11 , D13 , D15	Cross-references calendar intention (D11) with dedicated app usage (D01). Success is verified through cluster completions (D07), a surge in post-action efficiency ratings (D13), and reduced end-of-day cognitive exhaustion (D15).
A7: 5-Minute Friction Buster	D02 , D09 , D10 , D12 , D13	Captures initiation via doc creation (D02) and micro-compliance via continuous typing (D09) and generated content (D10) over 300 seconds. Goal achievement maps how successfully the burst pulverized high start-friction (D12) into clear flow (D13).
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Visual Relationship Cross-Reference

	D01	D02	D03
A1	X	X	
A2	X		
A3	X		X
A4	X		
A5	X		X
A6	X		
A7		X	
...			